$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p}, \tag{8.37}$$

$$\dot{\boldsymbol{p}} = \boldsymbol{F}(\boldsymbol{q}) - \frac{\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{F}(\boldsymbol{q})}{\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}}\boldsymbol{p}. \tag{8.38}$$

These equations preserve $\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} =$ constant. (Note: It is tempting to replace the denominator in this expression by K_0, the value of kinetic energy, but this is not recommended since it then destroys the first integral.) The system (8.37)–(8.38) can be rewritten in the alternative form

$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p},$$
$$\dot{\boldsymbol{p}} = [I - \mathcal{H}]\,\boldsymbol{F}(\boldsymbol{q}),$$

where $\mathcal{H} = \boldsymbol{p}(\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p})^{-1}\boldsymbol{p}^T\boldsymbol{M}^{-1}$ is a *projector* matrix which is idempotent: $\mathcal{H}^2 = \mathcal{H}$. The invariant subspace associated to this projector is span$(\boldsymbol{p})$:

$$\mathcal{H}\boldsymbol{p} = \boldsymbol{p}.$$

Isokinetic dynamics has nonvanishing divergence, which requires a special treatment to elucidate its statistical mechanical properties [338, 376]. In order to find a stationary solution, one substitutes

$$\rho(\boldsymbol{q},\boldsymbol{p}) = e^{-w(\boldsymbol{q},\boldsymbol{p})}\Phi(K(\boldsymbol{p})), \tag{8.39}$$

(with Φ a nonvanishing function of K) into the continuity equation for the isokinetic system, the function w being at this point unknown. Then one obtains, with $\boldsymbol{f}_{\text{iso}}$ denoting the vector field in (8.37)–(8.38), that

$$\begin{aligned} 0 = \nabla\cdot(\rho\boldsymbol{f}_{\text{iso}}) &= (\nabla\rho)\cdot\boldsymbol{f}_{\text{iso}} + \rho(\nabla\cdot\boldsymbol{f}_{\text{iso}}) \\ &= (-\nabla w\rho + \rho\frac{\Phi'(K)}{\Phi(K)}\nabla K)\cdot\boldsymbol{f}_{\text{iso}} + \rho(\nabla\cdot\boldsymbol{f}_{\text{iso}}). \end{aligned}$$

Using the isokinetic property, the term involving K drops out. One next calculates the divergence of the isokinetic vector field and finds that (see Exercise 5)

$$\nabla\cdot\boldsymbol{f}_{\text{iso}} = (\mathrm{N_c} - 1)\frac{\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{F}}{\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}}.$$

Thus the condition to have a stationary state of the form (8.39) is that

$$\nabla w \cdot \boldsymbol{f}_{\text{iso}} = (\mathrm{N_c} - 1)\frac{\boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{F}}{\boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{p}}.$$

A solution of this equation for w is easily found, it is

$$w = (\mathrm{N_c} - 1)\frac{U(\boldsymbol{q})}{K(\boldsymbol{p})}.$$

The invariant probability density, which is associated to a surface of constant kinetic energy, can be written as a generalized function using the Dirac notation:

$$\rho(\boldsymbol{q}, \boldsymbol{p}) = Z^{-1} e^{-(\mathrm{N_c}-1)U(\boldsymbol{q})/K(\boldsymbol{p})} \delta[K(\boldsymbol{p}) - K_0].$$

If we work on the surface of constant kinetic energy $K(\boldsymbol{p}) = K_0 \stackrel{\text{def}}{=} \beta^{-1}(\mathrm{N_c} - 1)$, the invariant density is of the form

$$\rho \propto \bar{\rho}_\beta \delta[K(\boldsymbol{p}) - \mathrm{k_B T}(\mathrm{N_c} - 1)]$$

which demonstrates that the isokinetic method, with this choice for the total kinetic energy, leaves invariant the configurational Gibbs distribution. Ergodicity has not been demonstrated, so it is not possible to conclude that the invariant distribution is unique; as we have seen before, the incorporation of stochastic terms can allow the formulation of strong statements regarding the ergodicity and mixing properties.

8.6.1 A Stochastic, Isokinetic Method for Gibbs Sampling

Minary et al. [271, 272] studied the isokinetic method as a means to control the instability in multiple timestepping due to resonance. They suggested that each physical degree of freedom be coupled to its own NHC thermostat through an isokinetic constraint in a "massive" thermostatting approach. These massive thermostats introduced some complexity (each degree of freedom had an associated Nosé-Hoover chain system as well as the constraint); obtaining an accurate solution for this large system is involving. A potentially more serious problem is that, although Nosé-Hoover chains and the isokinetic method may in some cases appear to be "practically ergodic," no proof of ergodicity exists. A recent article (by one of us, together with Margul and Tuckerman [220]) addressed these problems by replacing the Nosé-Hoover chains with stochastic terms (of Ornstein-Uhlenbeck type).